RTween API Reference

Complete API documentation for RTween system.

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Core API

RTween Class

Main static class for creating and controlling tweens.

Namespace

```csharp
using RG.Tweening;
```

Transform Tweening

Move

Animates a transform's world position.

```csharp
TweenHandle Move(Transform target, Vector3 endValue, float duration)
```

Parameters:

- `target` - The transform to animate
- `endValue` - Target world position
- `duration` - Animation duration in seconds

Returns: `TweenHandle` - Handle to control the tween

Example:

```csharp
RTween.Move(transform, new Vector3(5, 0, 0), If);
```

MoveX / MoveY / MoveZ

Animates a single axis of world position.

```csharp
TweenHandle MoveX(Transform target, float endValue, float duration)
TweenHandle MoveY(Transform target, float endValue, float duration)
TweenHandle MoveZ(Transform target, float endValue, float duration)
```

Parameters:

- `target` - The transform to animate
- `endValue` - Target position on specified axis
- `duration` - Animation duration in seconds

Example:

```csharp
```

```
RTween.MoveY(transform, lOf, 2f);
```

LocalMove

Animates a transform's local position.

```
csharp

TweenHandle LocalMove(Transform target, Vector3 endValue, float duration)
```

Example:

```
csharp

RTween.LocalMove(transform, Vector3.up *2, If);
```

LocalMoveX / LocalMoveY / LocalMoveZ

Animates a single axis of local position.

```
csharp

TweenHandle LocalMoveX(Transform target, float endValue, float duration)
TweenHandle LocalMoveY(Transform target, float endValue, float duration)
TweenHandle LocalMoveZ(Transform target, float endValue, float duration)
```

Rotation

Animates a transform's world rotation.

```
csharp

TweenHandle Rotation(Transform target, Quaternion endValue, float duration)
```

Example:

```
csharp

RTween.Rotation(transform, Quatemion.Euler(0, 180, 0), If);
```

LocalRotation

Animates a transform's local rotation.

```csharp
TweenHandle LocalRotation(Transform target, Quaternion endValue, float duration)
```

Scale

Animates a transform's local scale.

```csharp
TweenHandle Scale(Transform target, Vector3 endValue, float duration)
```

Example:

```csharp
RTween.Scale(transform, Vector3.one * 2, 0.5f);
```

ScaleX / ScaleY / ScaleZ

Animates a single axis of local scale.

```csharp
TweenHandle ScaleX(Transform target, float endValue, float duration)
TweenHandle ScaleY(Transform target, float endValue, float duration)
TweenHandle ScaleZ(Transform target, float endValue, float duration)
```

Jump

Animates a transform in an arc motion.

```csharp
TweenHandle Jump(Transform target, Vector3 endValue, float jumpPower, int numjumps, float duration)
```

Parameters:

- `target` - The transform to animate
- `endValue` - Target position
- `jumpPower` - Height of the jump
- `numjumps` - Number of jumps to perform
- `duration` - Total animation duration

Example:

```csharp
RTweenJump(transform, new Vector3(5, 0, 0), 2f, 3, 1.5f);
```

UI Tweening

UIMove

Animates a RectTransform's anchored position.

```csharp
TweenHandle UIMove(RectTransfonn target, Vector2 endValue, float duration)
```

Example:

```csharp
RectTransform rect = GetComponent<RectTransfonn>();
RTween.UIMovefrect, new Vector2(100, 50), If);
```

UIMoveX / UIMoveY

Animates a single axis of anchored position.

```csharp
TweenHandle UIMoveX(RectTransform target, float endValue, float duration)
TweenHandle UIMoveY(RectTransform target, float endValue, float duration)
```

SizeDelta

Animates a RectTransform's size delta.

```csharp
TweenHandle SizeDelta(RectTransform target, Vector2 endValue, float duration)
```

Example:

```csharp
RTween.SizeDelta(rect, new Vector2(200, 100), 0.5f);
```

Visual Tweening

Color

Animates color of UI or sprite components. Auto-detects: Image, Text, TextMeshPro, SpriteRenderer, Material.

```csharp
TweenHandle Color(Image target, Color endValue, float duration)
TweenHandle Color(Text target, Color endValue, float duration)
TweenHandle Color(TMP_Text target, Color endValue, float duration)
TweenHandle Color(SpriteRenderer target, Color endValue, float duration)
TweenHandle Color(Material target, Color endValue, float duration)
```

Example:

```csharp
Image image = GetComponent<Image>();
RTween.Color(image, Color.red, If);
```

Alpha

Animates alpha transparency. Auto-detects: Image, Text, TextMeshPro, SpriteRenderer, Material, CanvasGroup.

```csharp
```

```
TweenHandle Alpha(Image target, float endValue, float duration)
TweenHandle Alpha(Text target, float endValue, float duration)
TweenHandle Alpha(TMP_Text target, float endValue, float duration)
TweenHandle Alpha(SpriteRenderer target, float endValue, float duration)
TweenHandle Alpha(Material target, float endValue, float duration)
TweenHandle Alpha(CanvasGroup target, float endValue, float duration)
```

Example:

```csharp
CanvasGroup group = GetComponent<CanvasGroup>();
RTween.Alpha(group, 0f, 0.5f); // Fade out
```

Camera Tweening

FieldOfView

Animates camera field of view (perspective cameras).

```csharp
TweenHandle FieldOfView(Camera target, float endValue, float duration)
```

Example:

```csharp
RTween.FieldOfView(Camera.main, 90f, 2f);
```

OrthographicSize

Animates camera orthographic size (orthographic cameras).

```csharp
TweenHandle OrthographicSize(Camera target, float endValue, float duration)
```

Audio Tweening

Volume

Animates AudioSource volume.

```csharp
TweenHandle Vohime(AudioSource target, float endValue, float duration)
```

Example:

```csharp
AudioSource audio = GetComponent<AudioSource>();
RTween.Volume(audio, 0f, 1f); // Fade out
```

Pitch

Animates AudioSource pitch.

```csharp
TweenHandle Pitch(AudioSource target, float endValue, float duration)
```

Light Tweening

Intensity

Animates Light intensity.

```csharp
TweenHandle Intensity(Light target, float endValue, float duration)
```

Example:

```csharp
Light light = GetComponent<Light>();
RTween.lntensity(light, 5f, 1f);
```

Color (Light)

Animates Light color.

```csharp
TweenHandle Color(Light target, Color endValue, float duration)
```

Custom Values

Value

Animates a custom float value with callback.

```csharp
TweenHandle Value(float startValue, float endValue, float duration, Action<float> onUpdate)
```

Parameters:

- `startValue` - Starting value
- `endValue` - Target value
- `duration` - Animation duration
- `onUpdate` - Callback invoked each frame with current value

Example:

```csharp
RTween.Valuc(Of, lOOf, 2f, value =>
{
    Debug.Log($"Current: {value}");
    customSlider.value = value;
});
```

DelayedCall

Executes a callback after a delay.

```csharp
```

```
TweenHandle DelayedCall(float delay, Action callback)
```

Example:

```csharp
RTween.DelayedCall(2f, () => Debug.Log("Delayed!"));
```

Sequences

Sequence

Creates a new tween sequence for chaining animations.

```csharp
TweenSequence Sequence()
```

Sequence Methods

Add - Adds a tween to play after the previous one completes

```csharp
TweenSequence Add(TwcenHandlc tween)
```

Wait - Adds a delay

```csharp
TweenSequence Wait)float interval)
```

Call - Executes a callback

```csharp
TweenSequence Call(Action callback)
```

With - Adds a tween to play simultaneously with previous

```csharp
```

```
TweenSequence With(TweenHandle tween)
```

At - Adds a tween at specific time

```csharp
TweenSequence At(float time, TweenHandle tween)
```

Play - Starts the sequence

```csharp
TweenHandle Play()
```

Example:

```csharp
RTween.Sequence()
    .Add(RTween.Move(transform, Vector3.up * 2, 0.5f))
    .Wait(0.3f)
    .Add(RTween.Scale(transform, Vector3.one * 1.5f, 0.3f))
    .Call(() => Debug.Log("Done!"))
    .Play();
```

Effects

RTweenEffects Class

Static class for shake and punch effects.

Shake Effects

ShakePosition - Shakes world position

```csharp
TweenHandle ShakePosition(Transform target, float strength, float duration,
    int vibrato = 10, float randomness = 90f, bool fadeOut = true)
```

ShakeLocalPosition - Shakes local position

```csharp
```

```
TweenHandle ShakeLocalPosition(Transform target, float strength, float duration,
    int vibrato =10, float randomness = 90f, bool fadeOut = true)
```

ShakeRotation - Shakes rotation

```
csharp

TweenHandle ShakeRotation(Transform target, float strength, float duration,
    int vibrato = 10, float randomness = 90f, bool fadeOut = true)
```

ShakeScale - Shakes scale

```
csharp

TweenHandle ShakeScale(Transform target, float strength, float duration,
    int vibrato = 10, float randomness = 90f, bool fadeOut = true)
```

ShakeCamera - Shakes camera

```
csharp

TweenHandle ShakeCamera(Camera camera, float strength, float duration,
    int vibrato = 10, bool fadeOut = true)
```

ShakeUIPosition - Shakes UI element

```
csharp

TweenHandle ShakeUlPosition(RectTransform target, float strength, float duration,
    int vibrato = 10, float randomness = 90f, bool fadeOut = true)
```

Example:

```
csharp

RTweenEffects.ShakeCamera(Camera.main, 0.5f, 0.3f);
```

Punch Effects

PunchPosition - Punches world position

```
csharp
```

```
TweenHandle PunchPosition(Transform target, Vector3 punch, float duration,
    int vibrato =10, float elasticity =11)
```

PunchLocalPosition - Punches local position

```
csharp

TweenHandle PunchLocalPosition(Transform target, Vector3 punch, float duration,
    int vibrato = 10, float elasticity =11)
```

PunchRotation - Punches rotation

```
csharp

TweenHandle PunchRotation(Transform target, Vector3 punch, float duration,
    int vibrato = 10, float elasticity = If)
```

PunchScale - Punches scale

```
csharp

TweenHandle PunchScale(Transform target, Vector3 punch, float duration,
    int vibrato =10, float elasticity = If)
```

PunchUIPosition - Punches UI position

```
csharp

TweenHandle PunchUlPositionfRectTransform target, Vector2 punch, float duration,
    int vibrato = 10, float elasticity = If)
```

Example:

```
csharp

RTweenEffects.PunchScale(transfonn, Vector3.one * 0.2f, 0.31);
```

Quick Effects

QuickCameraShake - Quick camera shake

```
csharp
```

```
TweenHandle QuickCameraShake(Camera camera, float intensity = 0.3f)
```

QuickPunchScale - Quick scale punch

```
csharp

TweenHandle QuickPunchScale(Transform target, float intensity = 0.2f)
```

QuickUIShake - Quick UI shake

```
csharp

TweenHandle QuickUIShake(RectTransform target, float intensity = 10f)
```

Control Methods

TweenHandle Methods

SetEase - Sets easing function

```
csharp

TweenHandle SetEase(EaseType ease)
```

SetDelay - Sets start delay

```
csharp

TweenHandle SetDelay(float delay)
```

SetRepeat - Sets repeat behavior

```
csharp

TweenHandle SetRepeat(int count, RepeatType repeatType = RepeatType.Restart)
```

- `count`: Number of repeats (0 = no repeat, -1 = infinite)

OnStart - Sets start callback

```
csharp
```

```
TweenHandle OnStart(Action callback)
```

OnUpdate - Sets update callback

```csharp
TweenHandle OnUpdate(Action callback)
```

OnComplete - Sets completion callback

```csharp
TweenHandle OnComplete(Action callback)
```

Kill - Stops the tween

```csharp
void Kill(bool complete = false)
```

Pause - Pauses the tween

```csharp
void Pause()
```

Resume - Resumes the tween

```csharp
void Resume()
```

IsActive - Checks if tween is active

```csharp
bool IsActive { get; }
```

Example:

```csharp
```

```
TweenHandle handle = RTween.Move(transform, Vector3.up, If)
    .SetEase(EaseType.OutBounce)
    .SetDelay(0.5f)
    .SetRepeat(3, RepeatType.Yoyo)
    .OnComplete(() => Debug.Log("Complete!"));

// Later...
handle.Pause();
handle.Resume();
handle.Kill();
```

Global Control

Kill - Stops a specific tween

```
csharp

void Kill(TweenHandle handle, bool complete = false)
```

KillAll - Stops all active tweens

```
csharp

void KillAll()
```

Pause - Pauses a specific tween

```
csharp

void Pause(TweenHandle handle)
```

Resume - Resumes a specific tween

```
csharp

void Resume(TweenHandle handle)
```

PauseAll - Pauses all active tweens

```
csharp

void PauseAll()
```

ResumeAll - Resumes all active tweens

```csharp
void ResumeAll()
```

Enums

EaseType

Available easing functions:

```csharp
public enum EaseType
{
    Linear,
    InSine, OutSine, InOutSine,
    InQuad, OutQuad, InOutQuad,
    InCubic, OutCubic, InOutCubic,
    InQuart, OutQuart, InOutQuart,
    InQuint, OutQuint, InOutQuint,
    InExpo, OutExpo, InOutExpo,
    InCirc, OutCirc, InOutCirc,
    InBack, OutBack, InOutBack,
    InElastic, OutElastic, InOutElastic,
    InBounce, OutBounce, InOutBounce
}
```

RepeatType

Repeat behavior modes:

```csharp
public enum RepeatType
{
    Restart,     //Reset to start value each repeat
    Yoyo,        //Alternate between start and end
    Incremental  //Add the difference each repeat
}
```

Examples:

```csharp
//Restart: 0->1, 0->1, 0->1
RTween.MoveX(transform, 5f, 1f).SetRepeat(3, RepeatType.Restart);

// Yoyo: 0->I, 1->0, 0->1
RTween.MoveX(transform, 5f, 1f).SetRepeat(3, RepeatType.Yoyo);

//Incremental: 0->1, 1->2, 2->3
RTween.MovcX(transform, 5f, 1f).SetRepeat(3, RepeatType.Incremental);

// Infinite loop
RTween.MoveX(transform, 5f, 1f).SetRepeat(-1, RepeatType.Yoyo);
```

Complete Example

```csharp
```

```
using UnityEngine;
using RGames.Tweening;

public class CompleteExample : MonoBehaviour
{
    void Start()
    {
        //Basic tween with options
        RTween.Move(transform, Vector3.up * 5, 2f)
            .SetEase(EaseType.OutBack)
            .SetDelay(0.5f)
            ,OnComplete(() => Debug.Log("Movement done!"));

        // UI fade sequence
        CanvasGroup canvas = GetComponent<CanvasGroup>();
        RTween.ScqucnccO
            .Add(RTween.Alpha(canvas, Of, 0.5f))
            .Wait(lf)
            .Add(RTween.Alpha(canvas, If, 0.5f))
            .PlayO;

        //Infinite rotation
        RTween.Rotation(transform, Quatemion.Euler(0, 360, 0), 3f)
            .SetRepeat(-1, RepeatType.Incremental)
            .SetEase(EaseType.Linear);

        // Camera shake on hit
        RTweenEffects.QuickCameraShake(Camera.main, 0.5f);
    }
}
```